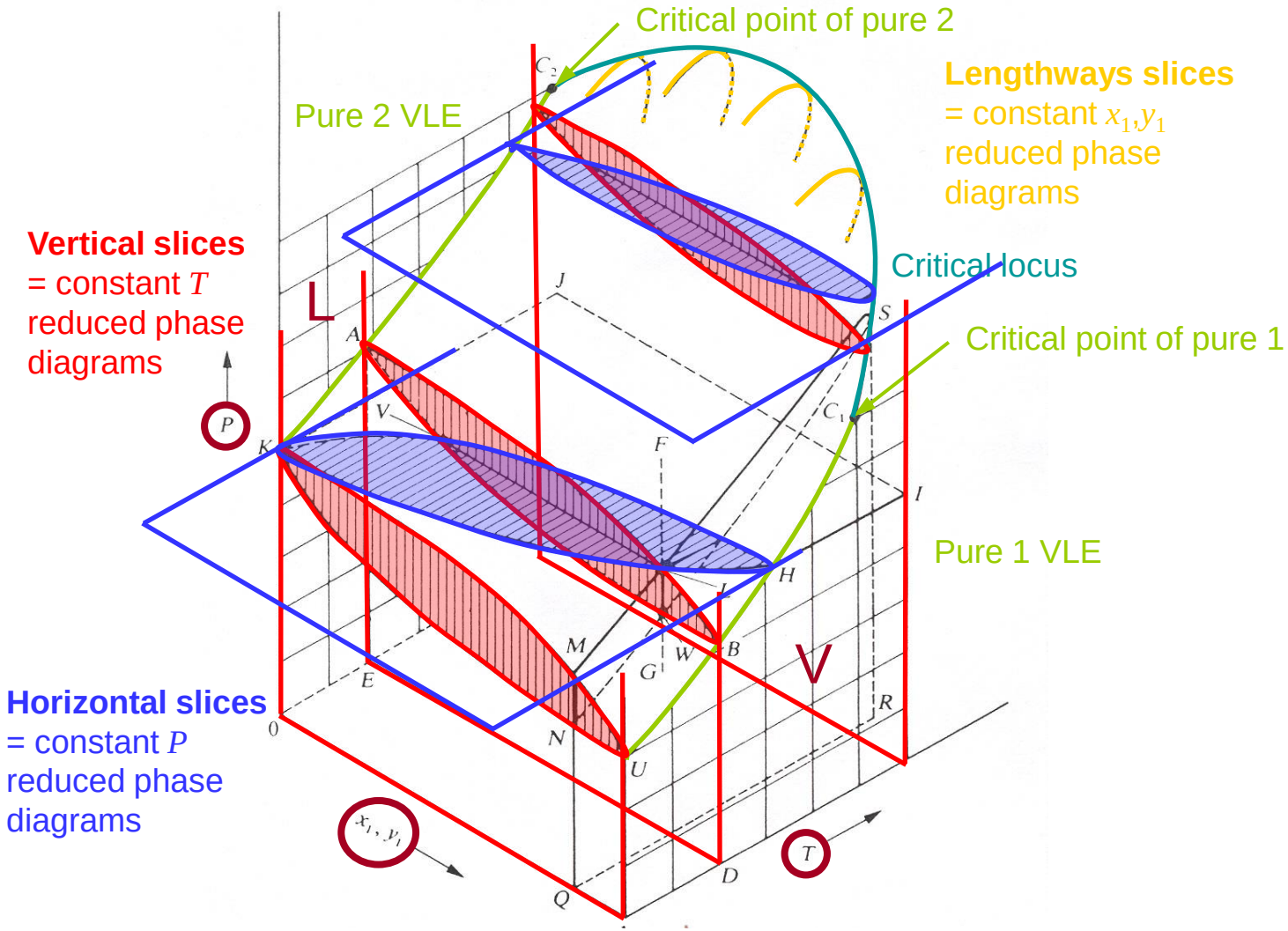
Recap – dealing with VLE

Binary mixtures have 3D phase diagrams

Reduced phase diagrams are 2D slices through 3D diagram



3D plot

VLE near critical points (using Equations of State)

If we are interested in VLE close to critical point(s) then we must use a single EOS to model both the vapour and the liquid

This is so that continuity is maintained

Unfortunately, not many substances can be accurately described by a single EOS for both the vapour and liquid

However mixtures of hydrocarbons, some inorganic gases and a few other substances can be dealt with by this approach.

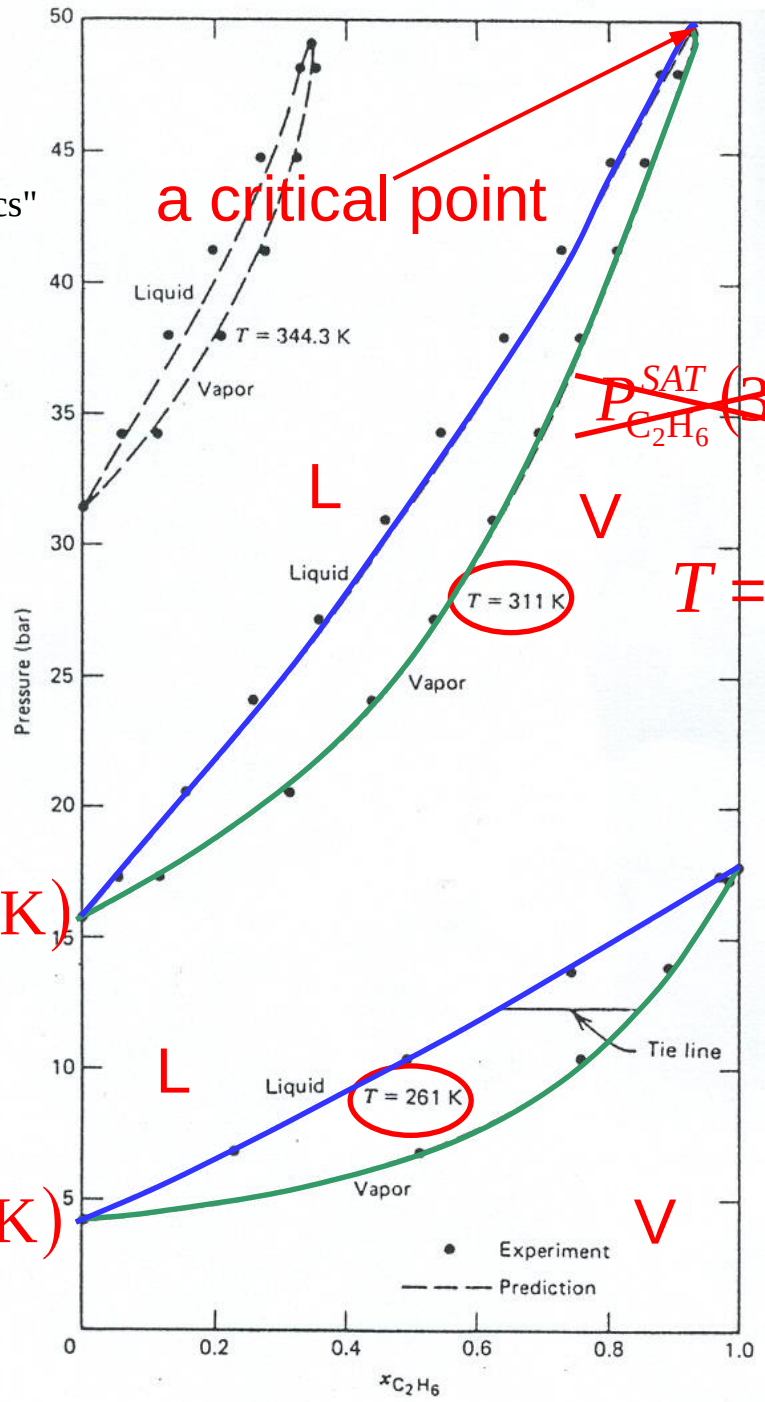
The starting point is, as usual,

$$\mu_i^l(P,T,x_1,x_2,.....) = \mu_i^v(P,T,y_1,y_2,.....)$$

or $\overline{f}_i^l = \overline{f}_i^v$

Calculations using a cubic EOS for the mixture, such as Peng-Robinson (see your Part I Process Calcs notes) are messy - we will examine some of the phenomena which occur near critical points and their practical implications

[Diagram taken from Sandler, "Chemical and Engineering Thermodynamics" , Wiley]



Constant temperature reduced phase diagrams for ethane-propene mixtures at three different temperatures

ethane T_C = 305.4K

propene T_C = 365K

~~$P^{SAT}_{C_2H_6}(311K)$~~

T = 311K is above T_C for ethane

$P^{SAT}_{C_2H_6}(261K)$

$P^{SAT}_{C_3H_6}(311K)$

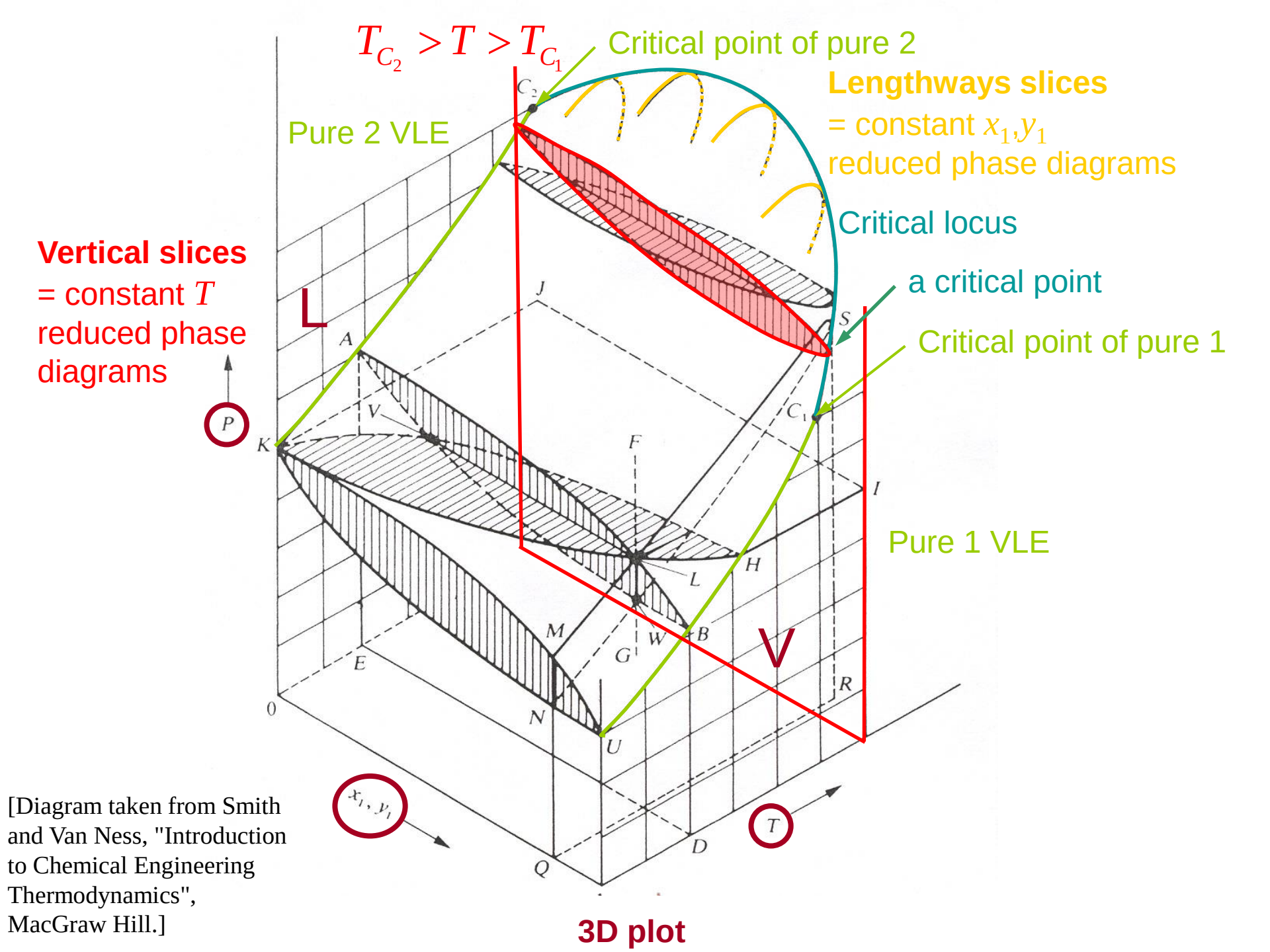
$P^{SAT}_{C_3H_6}(261K)$

Look first at T = 261K

Normal behaviour

Liquid line

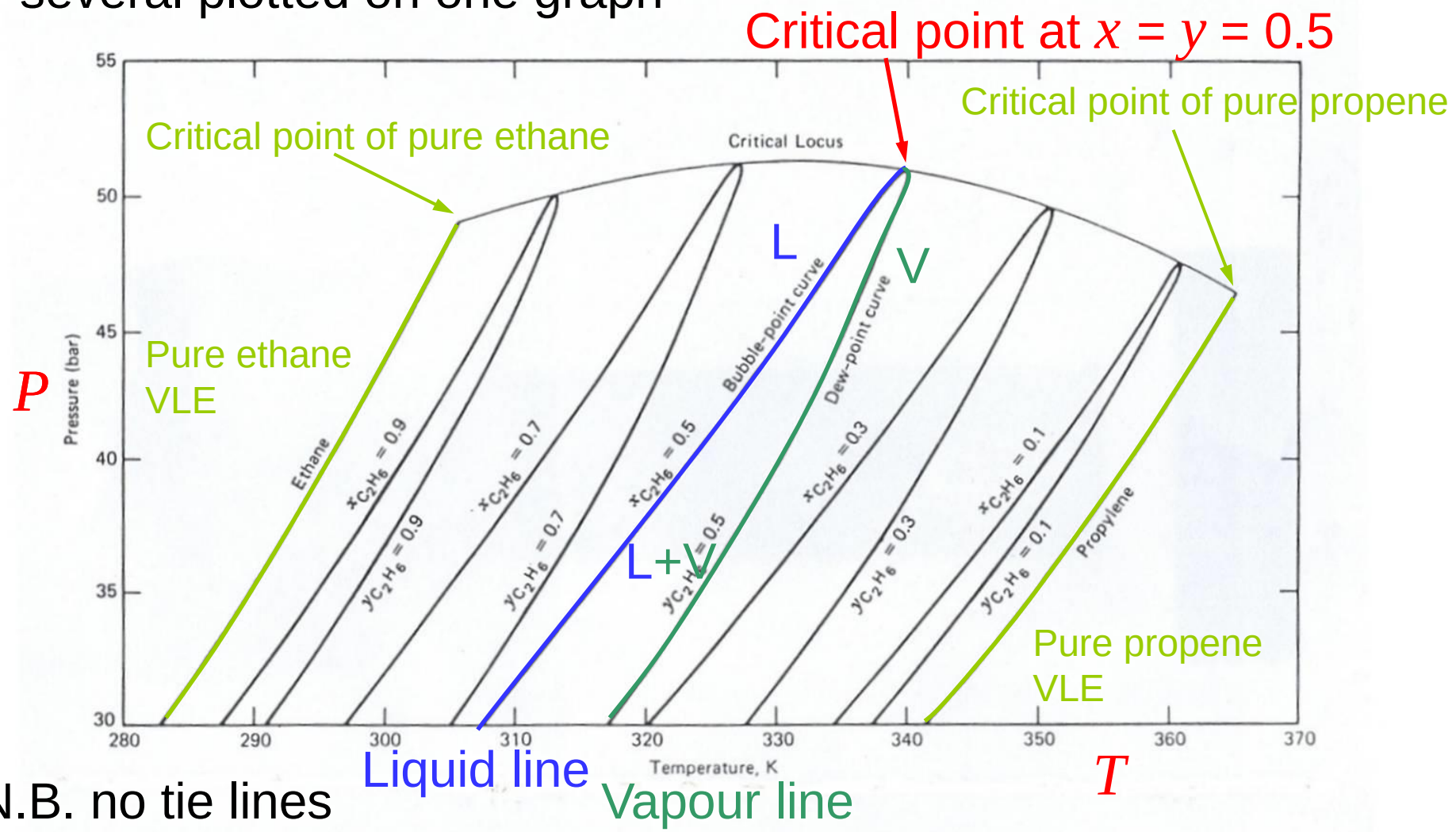
Vapour line



[Diagram taken from Smith and Van Ness, "Introduction to Chemical Engineering Thermodynamics", MacGraw Hill.]

3D plot

constant composition reduced phase diagram – several plotted on one graph



N.B. no tie lines

Liquid line

Vapour line

The bubble point, dew point and critical locus for the ethane-propene system

[Diagram taken from Sandler, "Chemical and Engineering Thermodynamics", Wiley]

Focus on one constant composition: $x,y = 0.5$

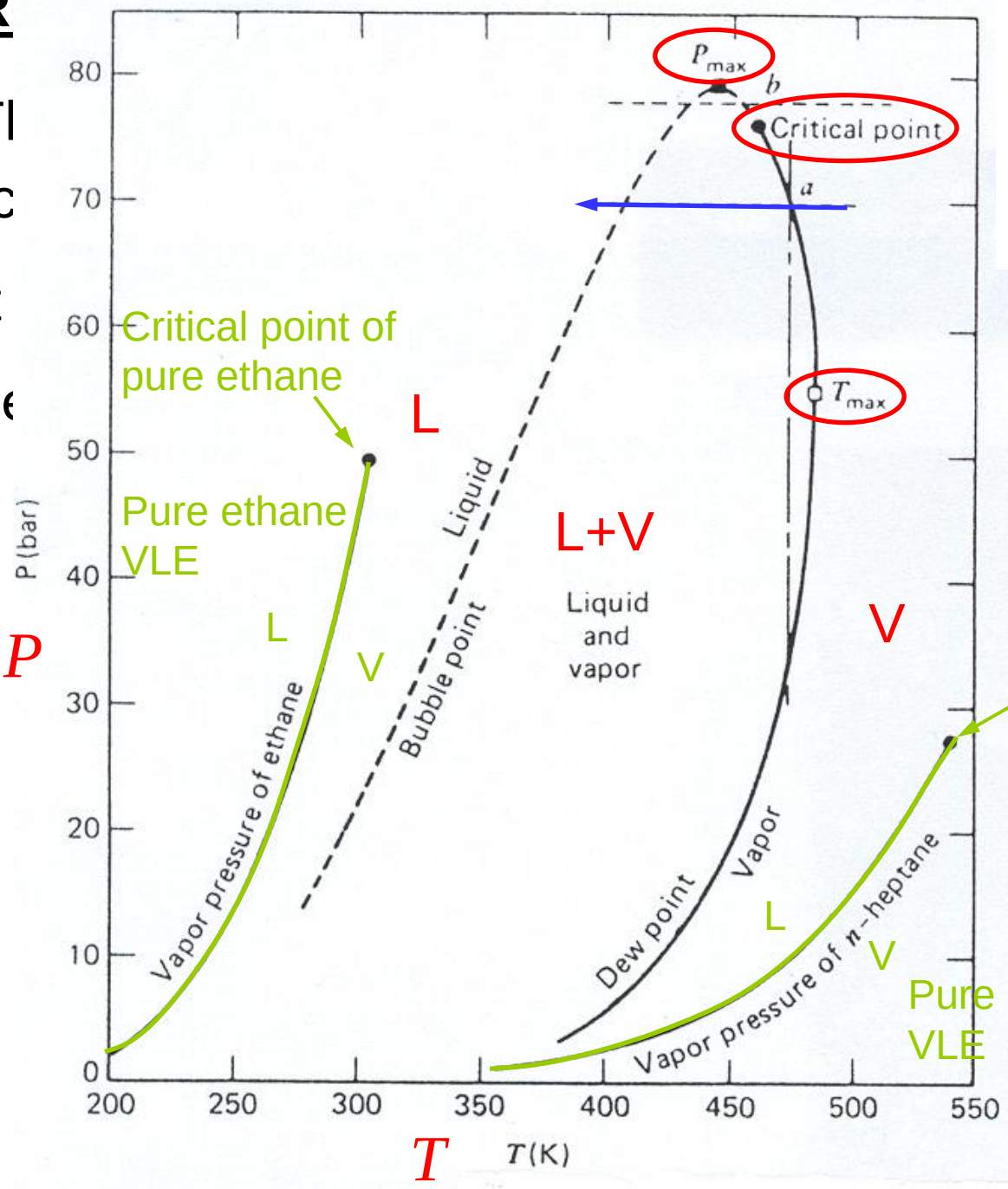
R

Tl

"c

It

Le



P

T

VLE for ethane, n-heptane and their constant composition mixture at 58.71 mol% ethane

= 58.71 mol% ethane

A reduced phase diagram at constant composition

her

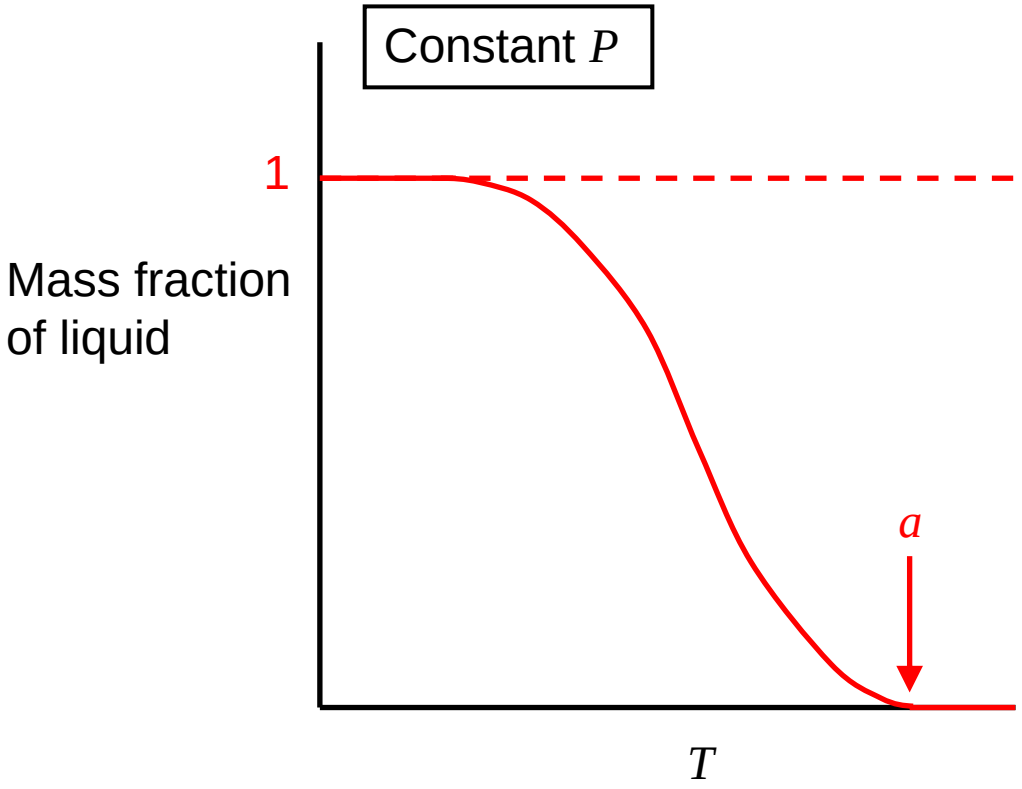
Consider (i)

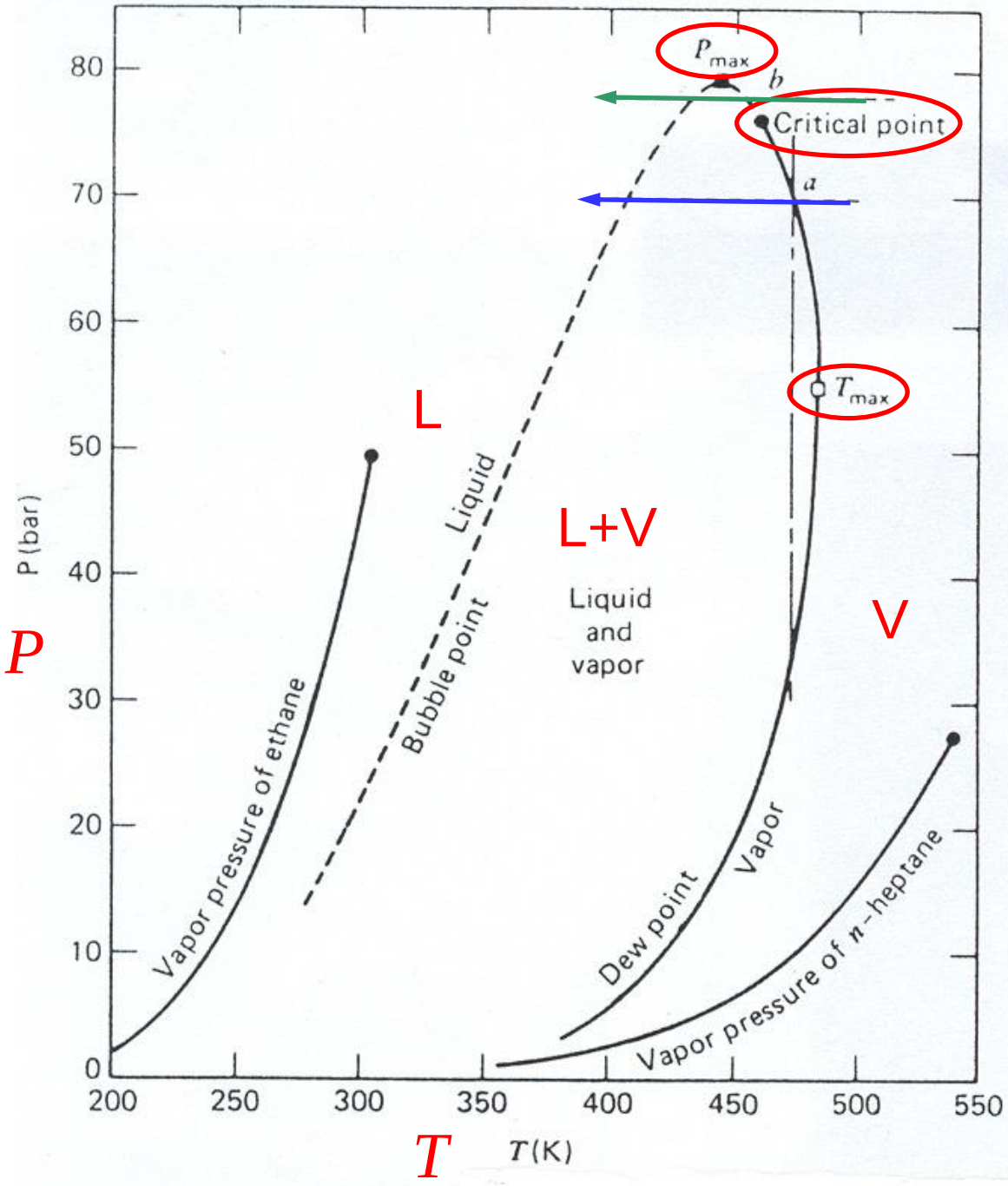
[Diagram taken from Sandler, "Chemical and Engineering Thermodynamics", Wiley]

(i) Starting from point a, reduce T at constant P:

Vapour $\rightarrow$ Liquid + Vapour $\rightarrow$ Liquid

(normal behaviour)





constant composition
= 58.71 mol% ethane

Consider (i)

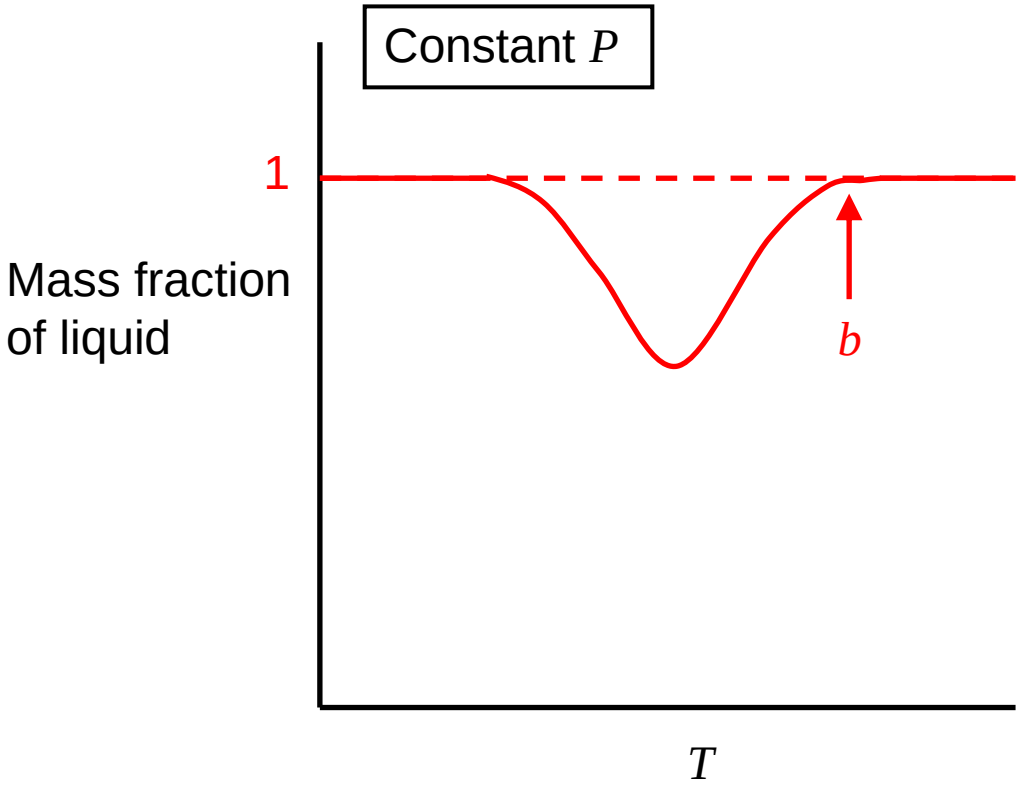
Consider (ii)

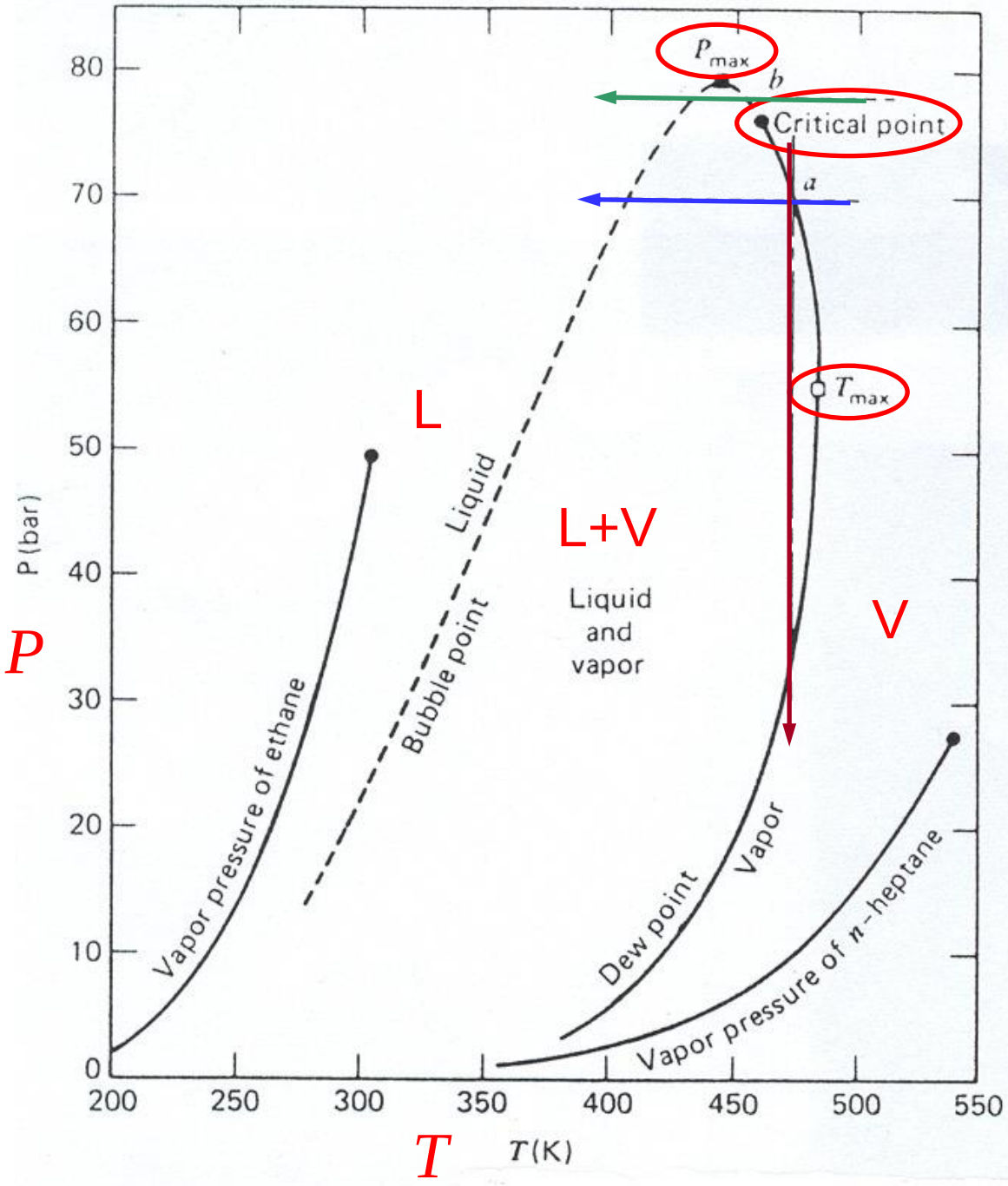
[Diagram taken from Sandler, "Chemical and Engineering Thermodynamics", Wiley]

(i) Starting from point b, reduce T at constant P:

Liquid $\rightarrow$ Vapour + Liquid $\rightarrow$ Liquid

(retrograde evaporation)





constant composition
= 58.71 mol% ethane

Consider (i)

Consider (ii)

Consider (iii)

[Diagram taken from Sandler, "Chemical and Engineering Thermodynamics", Wiley]

(i) Starting from point a, reduce P at constant T:

Vapour $\rightarrow$ Vapour + Liquid $\rightarrow$ Vapour

(retrograde condensation)

